Accounting for Empowerment?
Examining Women's Financial Inclusion in India
Online Appendix

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1 Financial inclusion in India

1.1 Bank branch and credit penetration

From 1969 until the early 1990s, several commercial banks were brought under the ownership and management of the government (Government of India, 1970). This increased the bank branch density significantly (International Monetary Fund, 1973). Between 1977 and 1990, a licensing policy by Reserve Bank of India (RBI) mandated that for every new branch opened in a location with existing bank infrastructure, four branches had to be opened in underserved areas (see Burgess and Pande (2005) for a description). The two policies increased the uptake of bank credit by rural households, and rural branch expansion corresponded with a decline in poverty (Burgess and Pande, 2005).

In the 1980s and early 1990s, policies aimed at increasing lending to priority sectors such as agriculture and socio-economically disadvantaged groups were introduced. The initiative by the National Bank for Agriculture and Rural Development (NABARD) to connect self-help groups (SHGs) with banks helped economically marginalized groups, especially women, access financial services[1]. These groups were initially identified through poverty estimates and later through community mapping processes[2]. These savings groups increased women's control over savings and income, and participation in borrowing decisions (Raghunathan et al., 2023; Kumar et al., 2021). They enabled peer groups that positively impacted women's self-esteem and personal agency (Basargekar, 2009; Kato and Kratze, 2013; Morgan and Coombes, 2013), and reduced experience of intimate partner violence (Goetz and Gupta, 1996; Rahman, 1999; Ahmed, 2005). However, the impact of SHGs varied in different parts of the country and some regions experienced higher non-performing assets (Sinha and Navin, 2021).

[1]This was implemented through the National Rural Livelihoods Project (NRLP) and corresponding state-specific Rural Livelihoods Mission.

[2]See RBI Master Circular RBI/2016-17/9: https://www.rbi.org.in/commonperson/English/Scripts/Notification.aspx?Id=1757

1.2 Early efforts to expand account ownership

Against this background, RBI announced the "no-frills account" policy in 2005, advising banks to offer accounts with low or no minimum balance requirements[3]. Banks were advised to offer these accounts and extend financial services in remote areas with the help of Business Correspondents (BCs)[4]. BCs particularly helped customers open accounts, make irregular and small transactions, process loans, and facilitate small deposits, remittances, insurance and pension products. In an attempt to scale up, BCs were expanded from non-profit organizations to include individuals[5]. They were allowed to conduct business for more than one bank and their operational area increased from 15 to 30 kms[6]. Despite these modifications, banking through BCs was unprofitable due to low volume of transactions (Uzma and Pratihari, 2019; Enclude and Grameen Foundation, 2013; Kolloju, 2014). Only 37% of all bank branches were located in rural areas, and 34% of the villages had access to banking services through BCs (Enclude and Grameen Foundation, 2013). The implementation suffered from technological issues, concerns about legal risks, and low financial literacy among clients (Khan, 2012; Enclude and Grameen Foundation, 2013). Only 58.7% of households reported having access to banking services[7] in the 2011 Population Census.

The no-frills account was subsequently restructured into the Basic Savings Deposit Account, offering digital services like ATMs and electronic payments[8]. This scheme led to an increase of 100 million accounts between 2011-13 as opposed to the 6 million through the introduction of BCs in 2007 (Helix Institute of Digital Finance (2015) cited in Uzma and Pratihari (2019)). Despite this, account ownership and access to bank infrastructure remained low.

1.3 Additional features of Pradhan Mantri Jan Dhan Yojana (PMJDY)

Anyone who opened the account in the first five months of the policy was eligible for life insurance as long as they were an income earning member of the household and between ages 18 and 59. Consistent with the policy, this covered only first time account holders with a debit card issued with the PMJDY account. Only one person in the household was eligible for this life insurance. There was no premium required under this policy. This INR 30,000 insurance coverage was available until financial year 2020. Over 125 million accounts were opened and over 110 million debit cards issued in the first five months of the policy. This provides a broad estimate of the people eligible for this insurance.

The policy also aimed to improve financial literacy, particularly in rural areas through Financial Literacy Centers set up by lead banks of districts or Centers of Financial Literacy that were set up at the block level by NGOs and sponsor banks. The literacy

[3] See RBI Master Circular RBI/2005-06/204)

[4] Full notification given in Circular RBI/2005-06/288

[5] Retired employees from banks, post office, government, teaching and other agents who worked with the government such as owners of fair price shops, insurance/ saving schemes agents.

[6] See RBI/2011-12/100

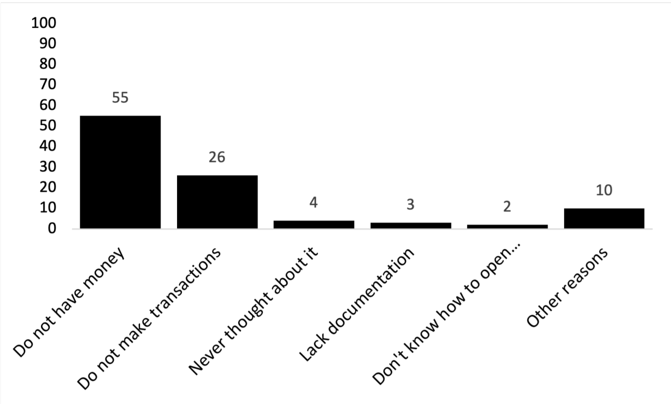
[7] These include services from brick and mortar bank, Business Correspondent and microfinance institution.

[8] See RBI notification RBI/2012-13/169

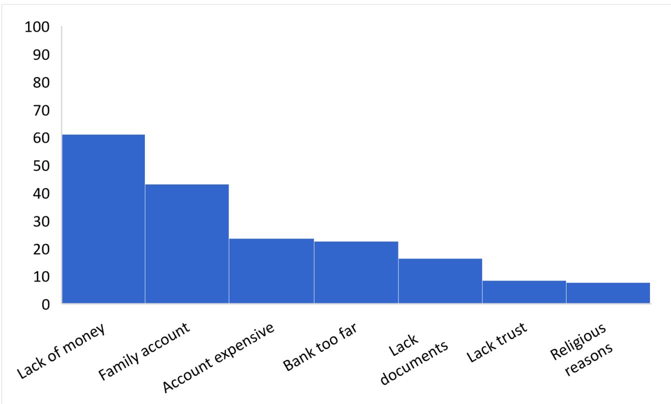
camps organized by these centers were free of charge. The RBI provides a list of these centers[9]. However, without information on when each center was opened and timing of the camps organized, it is difficult to estimate changes in financial literacy in the first year of PMJDY.

1.4 Reasons for not owning a bank account

Figure 1 reports reasons for adults (ages 18 and above) not having a bank account from two nationally representative surveys (Financial Inclusion Insights and World Bank Findex) . The most important reasons are not having sufficient money to store in an account, costs of owning a bank account (transactions and distance), and lack of documents required to open one.



Panel A: Financial Inclusion Insights, October 2013–January 2014



Panel B: World Bank Global Findex 2011

Figure 1: Reasons for not opening banka account before PMJDY

[9]Link: https://www.rbi.org.in/FinancialEducation/FLCs_CFLs_Details.aspx

1.5 Advancements in digital finance

This paper does not investigate the interaction of account ownership with digital infrastructure and financial products due to limited use of ATMs and mobile banking in 2014-15. Figure 2 shows that the annual transaction values through ATMs and mobile banking were a fraction (25% and 1%, respectively) of the value of total bank deposits in 2014. The low use of ATMs was explained by the limited machines in rural areas and significant transaction costs: only four free withdrawals were allowed per month. This paper does not capture the effects of the spike in mobile and internet banking transactions and short-term decline in ATM withdrawals (Figure 3) as a result of a monetary policy implemented in November 2016. This policy increased deposits in bank accounts, reduced cash supply and boosted e-wallet transactions (Chodorow-Reich et al., 2020). The analysis is restricted until 2016, determined by the timing of the post-policy survey data capturing women's decision making, where less than 0.007% of the survey observations were collected in November and December.

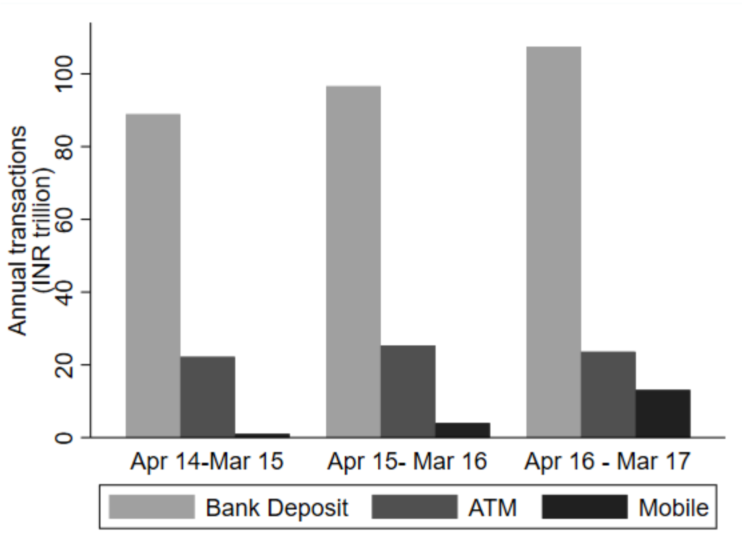


Figure 2: Annual volume of deposits versus digital transactions (2014-17)

Notes: Annual volume of deposits versus mobile and ATM transactions shows that a small fraction of the money held in savings accounts was transacted digitally between 2014 and 2017. The volume (in INR Trillion) is given in nominal terms. Source: Reserve Bank of India. Author's calculations.

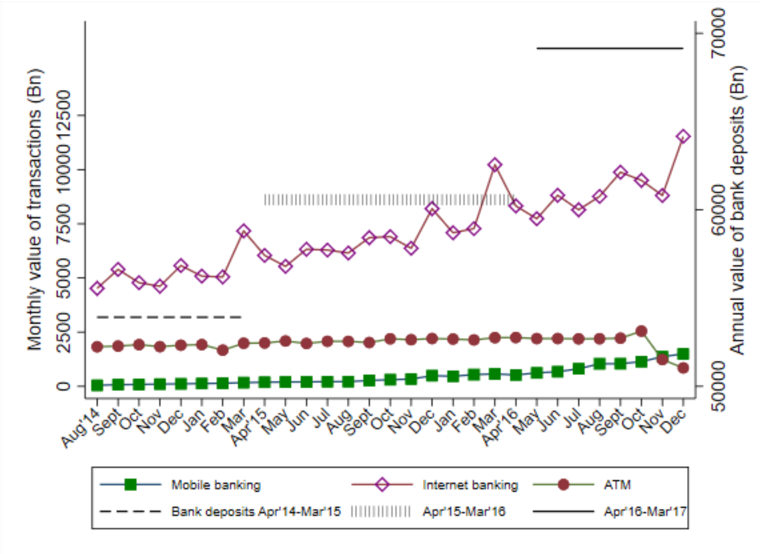


Figure 3: Monthly value of digital transactions in India (2014-16)

Notes: This figure shows the monthly volume of digital transactions (in INR Billion) in nominal terms by individuals using mobile phones, internet and ATMs. The horizontal lines in black (dash and solid) are the size of total bank deposits during that year. The figure identifies the effect of a nationwide change in monetary policy implemented in November 2016 (outside the analysis period). Source: Reserve Bank of India. Author's calculations.

2 Sampling methodology of household surveys

This section describes the sampling methodology of the three nationally representative household surveys used in the analysis.

Consumer Pyramids Household Survey (CPdx)

The CPdx stratifies 640 districts of the 2011 Population Census into 110 homogeneous regions (HRs) which groups districts with similar agro-climatic conditions, urbanization and female literacy levels as well as number of households. The North-Eastern states and Union Territories are each treated as a single HR. Each HR is further stratified based on population into a Rural stratum, a Very Large Towns stratum, a Large Towns stratum, a Medium-sized Towns stratum and a Small Towns stratum based on the number of households. The following regions were not fully surveyed: Andaman & Nicobar Islands, Arunachal Pradesh, Dadra & Nagar Haveli, Diu & Daman, Lakshadweep, Manipur, Meghalaya, Mizoram, Nagaland and Sikkim. The sampling strategy across the stratum is different: 25-30 villages were selected from each rural stratum using simple random sampling and at least 1 town from each town-size stratum (if available). Households within each village are selected using Systematic Random Sampling where every n^{th} household was selected (n ranging from 5 to 15) to survey a total of 16 households per village. For each selected town, 21 CEBs were randomly selected and 16 households were selected from each CEB using systematic random sampling. The survey, therefore, has a larger sample of urban households. There were 166,744 households (47,715 rural and 119,029 urban) in 438 districts surveyed in the first wave (January to April 2014) and 158,666 households (46,604 rural and 112,062 urban) in 425 districts in the fifth wave (May-August 2015). Every sample household within a strata is meant to represent the same number of households from the population using the survey weight. These weights are calculated using population projections by the survey implementers. This weight is generated for each round of survey.

India Human Development Survey (IHDS)

The survey was conducted in all states and union territories (except two - Andaman & Nicobar Islands and Lakshwadeep). It surveyed 384 out of 593 districts from the 2001 Census. The primary sampling units were villages and urban blocks. The number of urban blocks were selected using probability proportional to population. From each urban block, 15 households were surveyed. Half the rural sample comprised of 13,900 rural households from an older survey by the National Council of Applied Economics Research India. In each re-surveyed district, two additional villages were identified based on probability proportional to population, and 20 households were randomly included in the sample from these villages. The 2005 round covered 26,734 rural and 14,820 urban households while the 2012 round interviewed 27,579 rural and 14,573 urban households. Both samples were selected by stratified sampling. The second round in 2011-12 re-interviewed 83% of households in the first survey round.

Demographic Health Surveys

The survey uses 2011 Census information to draw a stratified two-stage sample. The primary sampling unit (PSU) is the village in rural areas and the census enumeration block (CEB) in urban areas. Villages are selected with probability proportional to size from the sampling frame. Within each stratum (rural/urban), six substrata are created based on number of households in village and percentage share of scheduled castes and scheduled tribes in a village. The PSUs were sorted by the literacy rate of women aged 6 and more, and final selection used probability proportional to size. Each selected urban and rural primary sampling unit was divided into segments of 100-150 households each and from each randomly selected segment, 22 households were randomly selected with systematic sampling.

2.1 Constructing district panel by 2001 Census boundaries

The DHS and RBI's administrative data is published using 2011 Population Census boundaries while both rounds of the IHDS are defined by the 2001 Census. Therefore, all districts in the DHS and RBI data were matched to their parent district in 2001. Districts which were formed from splitting of more than one parent district were excluded from the analysis to avoid misclassifications in the absence of population information at the time of split.

2.2 Summary statistics of sample

Table OA1: Descriptive statistics of household panel

	Wave 1 (Jan-Apr'14)		Wave 2 (May-Aug'14)		Wave 3 (Sep-Dec'14)		Wave 4 (Jan-Apr'15)		Wave 5 (May-Aug'15)	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Male head owns bank account (%)	95.08	(21.63)	100	(0)	100	(0)	100	(0)	99.99784	(0.47)
Wife of household head owns... (%)										
Bank account	0	(0)	0	(0)	23.54	(42.43)	23.55	(42.43)	24.22	(42.84)
Credit card	0.01	(0.8)	0.01	(1.)	0.43	(6.57)	0.15	(3.88)	0.11	(3.37)
Own mobile phone	12.33	(32.88)	13.57	(34.25)	22.72	(41.91)	24.45	(42.98)	28.47	(45.13)
Number of women in the household with...										
Bank account	0.04	(0.23)	0.05	(0.25)	0.31	(0.57)	0.33	(0.59)	0.37	(0.64)
Credit card	0.	(0.01)	0.	(0.01)	0.01	(0.08)	0.	(0.05)	0.	(0.05)
Own mobile phone	0.19	(0.45)	0.2	(0.47)	0.31	(0.55)	0.34	(0.57)	0.4	(0.62)
Number of men in the household with...										
Bank account	1.2	(0.59)	1.27	(0.56)	1.3	(0.59)	1.32	(0.61)	1.37	(0.65)
Credit card	0.01	(0.09)	0.01	(0.09)	0.03	(0.2)	0.02	(0.14)	0.02	(0.16)
Own mobile phone	1.21	(0.65)	1.27	(0.63)	1.31	(0.63)	1.33	(0.64)	1.36	(0.66)
Household demographics										
Household size	4.27	(1.33)	4.27	(1.33)	4.27	(1.34)	4.27	(1.34)	4.26	(1.35)
Number Adults	2.64	(0.97)	2.67	(1.)	2.71	(1.03)	2.75	(1.05)	2.8	(1.08)
Number daughters	0.73	(0.86)	0.71	(0.85)	0.7	(0.85)	0.68	(0.85)	0.66	(0.84)
Number children	1.63	(1.23)	1.59	(1.23)	1.56	(1.23)	1.53	(1.23)	1.47	(1.22)
Adult sex ratio	0.97	(0.38)	0.97	(0.4)	0.97	(0.41)	0.97	(0.42)	0.98	(0.44)
Real per-capita monthly total expenditure (INR)	1631.45	(959.58)	1663.37	(887.69)	1703.81	(845.69)	1741.13	(827.85)	1825.06	(900.04)

	Wave 1 (Jan-Apr'14)		Wave 2 (May-Aug'14)		Wave 3 (Sep-Dec'14)		Wave 4 (Jan-Apr'15)		Wave 5 (May-Aug'15)	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Urban (%)	94.31	(23.17)	94.19	(23.39)	94.17	(23.43)	94.09	(23.58)	94.02	(23.71)
Hindu (%)	82.58	(37.93)	82.47	(38.03)	82.49	(38.)	82.52	(37.98)	82.68	(37.85)
Muslim (%)	13.41	(34.08)	13.5	(34.17)	13.49	(34.17)	13.47	(34.15)	13.1	(33.74)
Christian (%)	0.74	(8.56)	0.66	(8.09)	0.66	(8.07)	0.66	(8.1)	0.78	(8.8)
Jain (%)	0.1	(3.16)	0.1	(3.23)	0.1	(3.23)	0.1	(3.16)	0.1	(3.2)
Buddhist (%)	0.32	(5.67)	0.33	(5.7)	0.31	(5.57)	0.32	(5.69)	0.31	(5.57)
Scheduled Caste (%)	23.64	(42.49)	23.83	(42.61)	23.82	(42.6)	23.83	(42.61)	23.97	(42.69)
Scheduled Tribe (%)	2.97	(16.99)	2.94	(16.88)	2.94	(16.89)	2.94	(16.9)	3.06	(17.23)
Other Backward Caste (%)	42.74	(49.47)	42.58	(49.45)	42.57	(49.45)	42.6	(49.45)	42.35	(49.41)
Intermediate Caste (%)	5.02	(21.83)	4.87	(21.53)	4.89	(21.57)	4.87	(21.52)	5.21	(22.23)
Upper Caste (%)	25.63	(43.66)	25.79	(43.75)	25.77	(43.74)	25.76	(43.73)	25.4	(43.53)
Male head: Age	44.01	(11.67)	44.31	(11.63)	44.63	(11.61)	44.92	(11.63)	45.4	(11.62)
Male head: Education	7.26	(5.31)	7.27	(5.33)	7.28	(5.33)	7.28	(5.32)	7.3	(5.3)
Wife: Age	38.93	(10.95)	39.23	(10.94)	39.56	(10.9)	39.86	(10.91)	40.31	(10.91)
Wife: Education	4.78	(4.94)	4.79	(4.96)	4.81	(4.97)	4.83	(4.97)	4.87	(4.96)
N	13148									

Notes: The table describes household characteristics of the sample using household weight provided by the CPDdx survey to reports statistics for five survey waves. Each wave includes four consecutive months. SD stands for standard deviation.

Table OA2: Descriptive statistics of district panel

	Before policy				After policy	
	Panel A: Individual and household covariates					
	IHDS 2005		IHDS 2011-12		DHS 2015-16	
	Mean	SD	Mean	SD	Mean	SD
Respondent has bank account	.16	(.37)	.36	(.48)	.53	(.5)
Household has bank account	.35	(.48)	.68	(.47)	.92	(.28)
Household size	5.66	(2.49)	5.51	(2.43)	5.45	(2.57)
Number Adults	3.04	(2.46)	3.05	(2.5)	3.51	(1.74)
Adult sex ratio	1.11	(.54)	1.16	(.6)	1.13	(.58)
Female headed household	.05	(.22)	.09	(.29)	.09	(.29)
Household in urban area	.32	(.47)	.31	(.46)	.3	(.46)
Household head: Hindu	.83	(.38)	.83	(.38)	.81	(.39)
Household head: Muslim	.12	(.33)	.13	(.33)	.14	(.35)
Household head: Christian	.02	(.14)	.02	(.12)	.02	(.14)
Household head: Sikh	.01	(.12)	.01	(.11)	.02	(.13)
Household head: Jain	.	(.05)	.	(.04)	.	(.04)
Household head: Buddhist	.01	(.08)	.01	(.09)	.01	(.1)
Head: Scheduled Caste	.23	(.42)	.23	(.42)	.22	(.41)
Head: Scheduled Tribe	.06	(.25)	.07	(.25)	.08	(.28)
Head: Other Backward Caste	.05	(.22)	.43	(.5)	.45	(.5)
Head: Education	5.35	(4.77)	5.63	(4.82)	6.21	(5.08)
Head: Age	32.64	(8.05)	47.16	(12.6)	46.96	(12.99)
Respondent: Education	4.08	(4.59)	5.15	(4.87)	6.18	(5.16)
Respondent: Age	33.33	(7.91)	33.9	(8.36)	32.95	(8.51)
Respondent is employed	.55	(.5)	.56	(.5)	.31	(.46)
Respondent: employed in agriculture	.45	(.5)	.37	(.48)	.16	(.37)
Respondent: employed non-agriculture sector	.08	(.27)	.17	(.38)	.14	(.34)

	Panel B: District-level banking			
	Mean	SD	Mean	SD
Total branches	725.54	(620.42)	1,099.98	(892.41)
Total accounts	1,517,502.72	(1,564,091.81)	3,156,879.08	(2,569,767.17)
Number districts		328		

	Panel C: Population Census			
	2001		2011	
	Mean	SD	Mean	SD
Rural (%)	74.6	(16.4)	70.76	(18.72)
Scheduled Caste/ Tribe (%)	24.57	(12.09)	25.81	(12.65)
Literate (%)	56.36	(11.3)	64.64	(9.81)
Electricity (hours)	.79	(.33)	.72	(.3)
Number of primary schools	1653.77	(893.85)	2065.74	(1212.89)
Number of middle schools	804.6	(482.53)	1425.1	(909.26)
Number of high schools	88.87	(60.04)	212.96	(157.96)
Number of colleges	28.05	(26)	71.12	(59.42)

Notes: Variables in panel A are population weighted aggregates from nationally representative household surveys, IHDS and DHS. The districts are defined by 2001 Population Census boundaries. Variables in panel B are district level aggregates generated using RBI's administrative census data. Variables in panel C are aggregates from Population Census 2001 and 2011. "Before Policy" includes mean estimates of the banking data from April 2006 through March 2014. "After Policy" includes April 2014 through March 2017.

3 Additional Figures

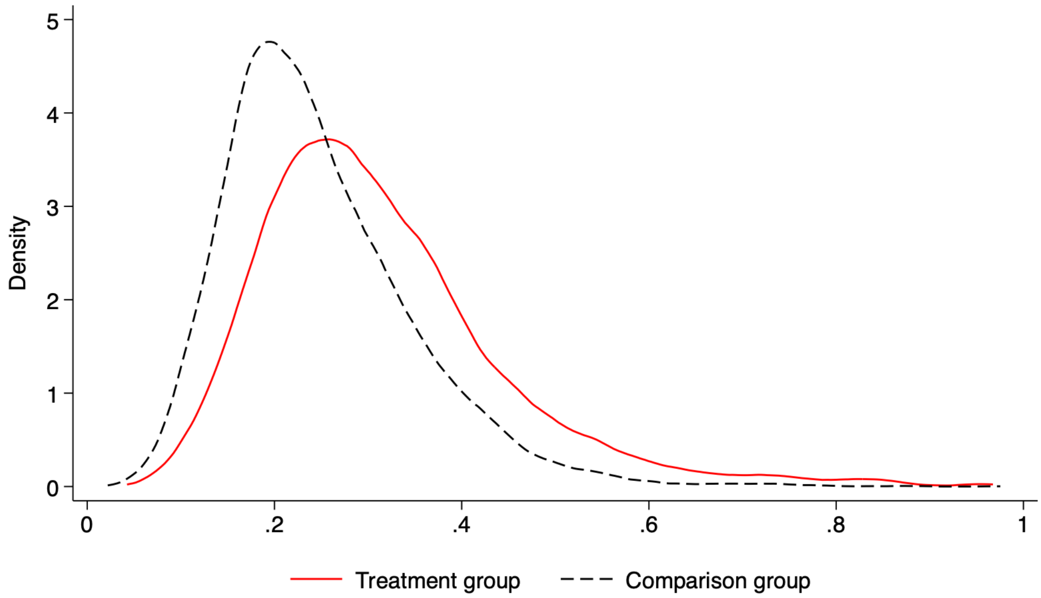


Figure 4: Distribution of Propensity Score across Treatment and Comparison Groups

Notes: The figure plots the predicted likelihood of household's assignment to treatment or comparison group based on household's observed characteristics before policy was implemented. The treatment group includes all households where the wife was unbanked before policy and opened a bank account in the survey wave after policy (September through December 2014). The comparison group includes households where the wife remained unbanked in the first year of the policy. Table 1 in the main paper describes the covariates included in predicting wives' propensity to be banked.

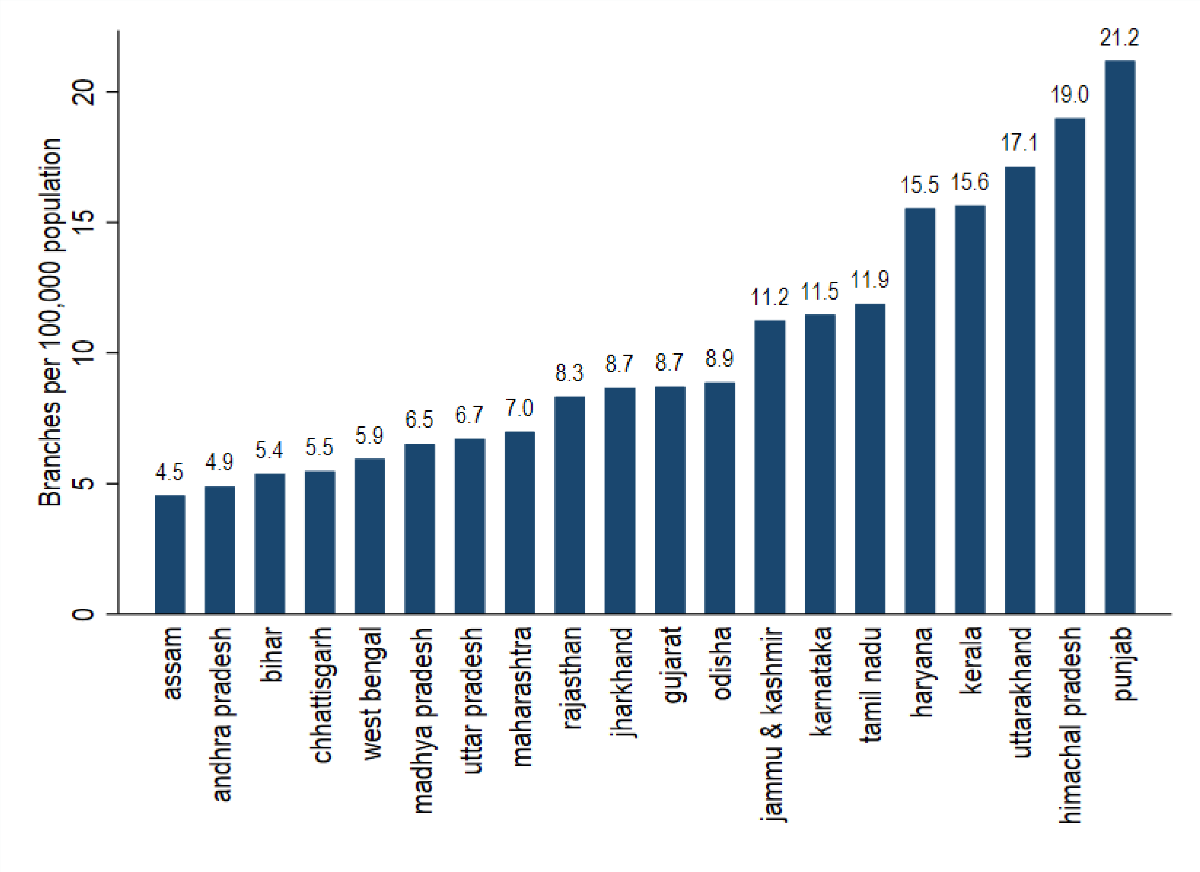


Figure 5: State-wise median bank branch density in 2014

Data sources: Basic Statistical Returns, Reserve Bank of India. Author's calculations.

4 Additional Tables

Table OA3: Monthly differences in consumption allocations between treated and comparison groups

	Women's consumption (1)	Cooking & sanitation (2)	Time saving (3)	Men's consumption (4)	Education (5)
Jan 2014	19.158***	-15.275***	0.119	7.268***	7.278***
	(3.782)	(5.799)	(0.183)	(2.708)	(1.796)
Feb 2014	9.475**	-12.565**	0.794***	9.657***	6.365***
	(3.696)	(5.749)	(0.192)	(2.687)	(1.777)
Mar 2014	0.629	-9.516	0.972***	11.637***	6.014***
	(3.658)	(5.841)	(0.184)	(2.783)	(1.799)
Apr 2014	10.391***	-11.757**	-0.033	13.682***	-2.822
	(3.448)	(5.847)	(0.202)	(2.767)	(1.779)
May 2014	1.672	-16.404***	0.404**	11.915***	-3.929**
	(3.468)	(5.067)	(0.161)	(2.467)	(1.756)
Jun 2014	-5.597	2.220	0.372**	10.790***	3.804**
	(3.423)	(4.384)	(0.148)	(2.143)	(1.740)
Jul 2014	-5.868*	-0.236	-0.062	7.104***	5.919***
	(3.092)	(3.274)	(0.154)	(1.484)	(1.636)
Sep 2014	-3.463	-2.093	-0.623***	1.219	-3.779***
	(3.256)	(3.208)	(0.150)	(1.275)	(1.293)
Oct 2014	21.221***	-1.684	-0.294**	4.871***	-2.060
	(3.387)	(3.853)	(0.141)	(1.691)	(1.300)
Nov 2014	3.034	-16.417***	-0.500***	4.014*	-4.186***
	(3.369)	(4.854)	(0.161)	(2.158)	(1.460)
Dec 2014	-7.997**	-27.735***	-0.163	2.855	-5.538***
	(3.236)	(5.292)	(0.151)	(2.485)	(1.592)
Jan 2015	-1.855	-24.569***	-0.555***	1.964	-3.613**
	(3.399)	(5.425)	(0.164)	(2.485)	(1.640)
Feb 2015	-12.244***	-29.533***	-0.315**	3.782	-6.205***
	(3.219)	(5.476)	(0.155)	(2.633)	(1.634)
Mar 2014	-6.013*	-32.715***	-0.678***	8.287***	-10.828***
	(3.503)	(5.686)	(0.163)	(2.722)	(1.741)
Apr 2015	-15.792***	-28.833***	-0.431***	11.641***	-14.035***
	(3.436)	(5.732)	(0.152)	(2.760)	(1.747)
May 2015	-5.304	-25.073***	-0.505***	11.316***	-15.218***
	(3.361)	(5.880)	(0.173)	(2.884)	(1.737)
Jun 2015	-8.330**	-18.149***	-0.750***	10.063***	-7.229***
	(3.512)	(6.150)	(0.183)	(2.910)	(1.874)
Jul 2015	0.724	-4.308	-1.010***	14.340***	-5.759***
	(3.689)	(6.417)	(0.189)	(3.011)	(1.969)
Aug 2015	-9.320**	6.262	-0.702***	12.147***	-8.852***
	(3.798)	(6.451)	(0.195)	(3.052)	(1.939)
Observations	256542	256542	256542	256542	256542
Covariates	Yes	Yes	Yes	Yes	Yes

Notes: Expenditure categories listed as column titles. Standard errors are reported in parentheses and clustered at household level. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table OA4: Differences in miscellaneous consumption categories between treatment and comparison groups by survey-wave

	Appliances	Restaurants	Entertainment	Rent & Utilities	Transport	Communication	Health
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Per capita monthly expenditure in real terms							
Jan-Apr14	-0.915**	-0.652	0.140	-0.020	6.212***	1.949**	1.239
	(0.383)	(1.149)	(0.206)	(0.370)	(1.230)	(0.912)	(0.801)
Sep-Dec14	-1.693***	-2.012*	0.232	1.457***	-9.885***	-0.602	1.936**
	(0.375)	(1.214)	(0.209)	(0.391)	(1.231)	(0.919)	(0.799)
Jan-Apr15	-1.926***	-2.690**	0.040	1.756***	-13.721***	-3.951***	1.387*
	(0.382)	(1.246)	(0.182)	(0.393)	(1.259)	(0.918)	(0.831)
May-Aug15	-2.374***	2.349*	0.696***	1.942***	-14.730***	-3.913***	2.925***
	(0.394)	(1.270)	(0.202)	(0.403)	(1.261)	(0.983)	(0.832)
Panel B: Share of total monthly consumption							
Jan-Apr14	-0.000*	0.000	0.000**	0.003***	0.002***	0.001*	-0.000
	(0.000)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)	(0.000)
Sep-Dec14	-0.000**	-0.001**	0.000*	-0.005***	0.000	0.002***	0.001***
	(0.000)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)	(0.000)
Jan-Apr15	-0.001***	-0.002***	-0.000	-0.006***	0.000	0.002***	0.001***
	(0.000)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)	(0.000)
May-Aug15	-0.001***	0.001	0.000*	-0.007***	0.000	0.002***	0.001***
	(0.000)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)	(0.000)
Observations	48165	48165	48165	48165	48165	48165	48165
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports estimated differences in per capita monthly consumption allocations (panel A) and share of total monthly consumption (panel B) between the treatment and comparison groups by survey wave from the augmented inverse-probability weighted DiD. The first row provides evidence of parallel trends before policy. The second survey wave, May through August is the reference group. The dependent variables are household's per capital monthly expenses in real terms and specified in column titles. Column 1 includes kitchen, household and mobile appliances; column 2 includes expenditure on food and drinks in restaurants, canteens and food stalls; column 3 includes tickets for movies, theatre, music concerts, sport events and other amusement activities; column 4 includes house rent and utilities; column 5 consists of modes of transport; column 6 includes phones, TV, internet and other media. Column 7 includes expenditure on medicines, doctor's/physiotherapists and hospitalization fees, medical tests and health insurance. The sample is restricted to households where the wife of household head didn't have a bank account in the second survey wave. The treatment group includes households where the female spouse of household head opened a bank account in survey wave 3 (September - December 2014) and comparison group includes households where the spouse didn't own a bank account at least 1 year after implementation (waves 1-5). The outcome and treatment models include covariates listed in Table 1 of the paper. The estimation includes household fixed effects. Standard errors are clustered by household and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table OA5: Differences in expansion of account ownership between High and Low Impact districts

Dependent variable: Percent difference in accounts since 2014		
	(1)	(2)
High impact × 2006	-0.282	-0.664
	(2.238)	(2.039)
High impact × 2007	0.394	0.064
	(2.236)	(2.032)
High impact × 2008	0.198	-0.221
	(2.281)	(2.084)
High impact × 2009	0.521	0.219
	(2.277)	(2.075)
High impact × 2010	0.941	0.666
	(2.319)	(2.130)
High impact × 2011	2.633	2.407
	(2.327)	(2.137)
High impact × 2012	3.398	3.280
	(2.205)	(2.091)
High impact × 2014	4.624**	4.544**
	(2.087)	(1.949)
High impact × 2015	7.622***	7.802***
	(2.547)	(2.277)
High impact × 2016	11.660***	12.205***
	(3.360)	(2.726)
High impact × 2017	13.918***	14.793***
	(4.185)	(3.280)
Observations	3156	3132
R^2	0.921	0.952
Adjusted R^2	0.892	0.928
District FE	Yes	Yes
Time FE	Yes	Yes
State*Year FE	No	Yes
LASSO controls	Yes	Yes

Notes: The table reports year-by-year differences in bank branch and account variables between High and Low Impact districts. High impact includes districts at state median bank branch density and below. Low impact includes the districts above state median. The dependent variables is percent growth of bank accounts since 2014. The reference group is financial year 2013. The estimation follows Eq. 1 in main text and includes district, year and state*year fixed effects and controls selected by post double (LASSO) selection method (Belloni et al., 2013). Standard errors are reported in parentheses and clustered at district level. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table OA6: Parallel trends test: Other empowerment outcomes (1/2)

	Employed		Fertility		Mobility outside home	
	(1)	(2)	(3)	(4)	(5)	(6)
High impact × Pre-treatment time dummy	0.014 (0.020)	0.004 (0.013)	0.059* (0.032)	0.061** (0.024)	-0.000 (0.019)	-0.002 (0.017)
Observations	37663	37663	36484	36484	51269	51269
R^2	0.251	0.288	0.496	0.512	0.046	0.110
Control group mean	0.522	0.522	2.434	2.434	0.164	0.164
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
State*Year FE	No	Yes	No	Yes	No	Yes
LASSO controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports differences in empowerment outcomes between High and Low Impact districts in the years before policy implementation. Dependent variables are listed as column titles - respondent is employed, women's fertility (number of children), and ability to visit friends/relatives and public spaces on her own. High impact includes districts with bank branch density equal to the state median or less. Low impact includes the districts with branch density greater than state median. Bank branch density is calculated per 100,000 population. Only coefficients of interaction terms of High Impact districts with year dummy are reported. Districts are defined by 2001 Population Census Boundary. The sample includes districts where the account expansion policy (PMJDY) was implemented from August 2014 to August 2015. Decision making variables are extracted from nationally representative household surveys (IHDS 2005 and 2012), bank branch density is estimated using data from RBI and Population Census. All specifications include district and year fixed effects and controls selected by post double (LASSO) selection method (Belloni et al., 2013). Columns 2, 4 and 6 include state × year fixed effects. Standard errors are clustered by district and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table OA7: Parallel trends test: Other empowerment outcomes (2/2)

	Cruelty by spouse/ his family	
	(1)	(2)
High impact × (Year=2012)	-5.957	-6.185
	(12.635)	(12.823)
High impact × (Year=2013)	-16.608	-17.202
	(14.651)	(13.522)
Observations	923	920
R^2	0.953	0.960
Control group mean	190.43	190.43
District FE	Yes	Yes
Time FE	Yes	Yes
State*Year FE	No	Yes

Notes: This table reports differences in empowerment outcomes between High and Low Impact districts in the years before policy implementation. Dependent variables is listed as column title - annual dowry-related violence cases in a district. High impact includes districts with bank branch density equal to the state median or less. Low impact includes the districts with branch density greater than state median. Bank branch density is calculated per 100,000 population. Only coefficients of interaction terms of High Impact districts with year dummy are reported. Districts are defined by 2001 Population Census Boundary. The sample includes districts where the account expansion policy (PMJDY) was implemented from August 2014 to August 2015. Annual crimes are extracted from the National Crime Records Bureau, bank branch density is estimated using data from RBI and Population Census. All specifications include district and year fixed effects. Column 2 includes state × year fixed effects. Standard errors are clustered by district and reported in parentheses. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table OA8: Aggregate treatment effect: Wife's account ownership and share of consumption allocations

Dependent variable: Share of per capita monthly expenditure

	Women's consumption	Cooking & sanitation	Time saving	Men's consumption	Education
Post × Treatment	0.005***	0.011***	-0.0004***	-0.002	-0.006***
	(0.002)	(0.003)	(0)	(0.001)	(0.001)
	[0.003]	[0]	[0]	[0.21]	[0]
Sharpened q-value	0.002	0.001	0.001	0.044	0.001
Comparison group mean	0.058	0.623	0.	0.073	0.028
Observations	48164	48164	48164	48164	48164
Covariates	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes
Joint significance of differences (pre-policy)					
Chi-squared value	2.116	20.211	12.579	3.264	0.115
p-value	0.146	0	0	0.071	0.734

Notes: This table reports the effect of wife's account ownership on household's consumption using an augmented inverse probability weighted DiD. The first row reports aggregate treatment effect. The treatment group includes households where female spouse of household head opened a bank account in survey wave 3 (September - December 2014) and comparison group includes households where the spouse didn't own a bank account at least 1 year after implementation. The post-treatment dummy variable is assigned 1 for CPdx's surveys 3-5 (September 2014 to August 2015) and 0 for survey waves 1 and 2 (January to August 2014). The dependent variables are household's share of per capita monthly consumption and are listed as column titles. Column 1 includes household's expenditure on clothing and footwear, cosmetics, accessories, and beauty goods and services; column 2 includes items complementary to domestic chores of cooking and cleaning such as food, utensils and toiletries; column 3 includes time saving appliances (kitchen appliances) and services (domestic help). Column 4 includes expenditure on tobacco products and liquor, shaving articles and fuel for purposes other than cooking. Column 5 analyzes total expenditure on education such as fees to schools, colleges/ private tuition, books, uniforms etc. The second row includes standard errors in parentheses that are clustered by household. The third row reports p values in square brackets and fourth row reports sharpened two- stage q-values that correct the p-value of the interaction coefficient for false discovery rate from testing multiple hypotheses. The last two rows are a test for joint significant of differences between treatment and comparison groups before policy reporting the chi-squared statistic and corresponding p value. The sample is restricted to households where the wife of household head didn't have a bank account in the second survey wave.The estimation controls for time varying characteristics of households and total monthly consumption expenditure. It also includes household and survey wave fixed effects. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

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